

Shifan Xu

PH.D. CANDIDATE IN APPLIED PHYSICS

Yale University, New Haven, CT

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Education

Yale University

PH.D. CANDIDATE IN APPLIED PHYSICS

New Haven, CT, USA

Sep. 2021 – Present

- Advisors: Prof. Yongshan Ding and Prof. Steven M. Girvin

University of Chicago

VISITING PH.D. STUDENT

Chicago, IL, USA

Jan. 2024 – May 2024

Nanjing University

B.S. IN PHYSICS

Nanjing, China

Sep. 2017 – Jun. 2021

University of California, Berkeley

VISITING UNDERGRADUATE STUDENT

Berkeley, CA, USA

Jul. 2019 – Jan. 2020

Research Experience

Yale University

DOCTORAL RESEARCH; GRADUATE RESEARCH ASSISTANT, 2023–PRESENT

New Haven, CT

Sep. 2021–Present

- **Quantum memory architecture.** Developed system-level and high-bandwidth shared QRAM architectures, with first-author papers at MICRO 2023 and ASPLOS 2025; developed QRAM simulators.
- **Fault-tolerant quantum architecture.** Developed magic-state distillation for the bicycle architecture (first author, ISCA 2026); contributed to heterogeneous error correction, qLDPC routing, and simultaneous magic-state injection.
- **Error-corrected QRAM.** Ongoing research on fault-tolerant quantum memory queries; manuscript in preparation.
- **Algorithm resource estimation.** Developing full-stack resource-estimation methods for quantum algorithms on neutral-atom qLDPC systems, linking DF-THC compilation, logical-qubit mapping, and magic-state delivery; preliminary FeMoco estimates.
- **Emerging direction.** Exploring AI methods for quantum architecture design.

Pacific Northwest National Laboratory

GRADUATE STUDENT INTERN; MENTOR: ANG LI

Richland, WA

Jun.–Aug. 2025

- Researched secure quantum queries, leading to the co-first-author OQRAM preprint on oblivious QRAM for securing delegated quantum queries.

Publications

PEER-REVIEWED PUBLICATIONS AND ACCEPTED PAPERS

- Shifan Xu**, Kun Liu, Patrick Rall, Zhiyang He, and Yongshan Ding. “Distilling Magic States in the Bicycle Architecture.” In Proceedings of the 2026 ACM/IEEE 53rd Annual International Symposium on Computer Architecture (**ISCA**), pp. 2081–2095. 2026. DOI: 10.1109/ISCA66397.2026.00148 · arXiv:2602.20546.
- Kun Liu, Takahiro Tsunoda, Sophia H. Xue, Evan McKinney, Zeyuan Zhou, **Shifan Xu**, Robert J. Schoelkopf, and Yongshan Ding. “Efficient Routing of Quantum LDPC Codes on Programmable 2D Toric Architectures.” Accepted to the IEEE International Conference on Quantum Computing and Engineering (**QCE**), 2026. arXiv:2604.18714.
- Shifan Xu**, Alvin Lu, and Yongshan Ding. “Fat-Tree QRAM: A High-Bandwidth Shared Quantum Random Access Memory for Parallel Queries.” Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), Volume 2. 2025. DOI: 10.1145/3676641.3716256 · arXiv:2502.06767.
- Samuel Stein, **Shifan Xu**, Andrew W. Cross, Theodore J. Yoder, Ali Javadi-Abhari, Chenxu Liu, Kun Liu, Zeyuan Zhou, Charles Guinn, Yufei Ding, Yongshan Ding, and Ang Li. “HetEC: Architectures for Heterogeneous Quantum Error Correction Codes.” Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), Volume 2. 2025. DOI: 10.1145/3676641.3716001 · arXiv:2411.03202.
- Shifan Xu**, Connor T. Hann, Ben Foxman, Steven M. Girvin, and Yongshan Ding. “Systems Architecture for Quantum Random Access Memory.” In Proceedings of the 56th Annual IEEE/ACM International Symposium on Microarchitecture (**MICRO**), pp. 526–538. 2023. DOI: 10.1145/3613424.3614270 · arXiv:2306.03242.

PREPRINTS

- Shifan Xu**^{*}, Yizhuo Tan^{*}, Yongshan Ding, and Jakub Szefer. “OGRAM: Oblivious Quantum Random Access Memory for Securing Delegated Quantum Queries.” arXiv preprint arXiv:2607.27171 (2026). ^{*}Co-first authors.
- Dantong Li, **Shifan Xu**, and Yongshan Ding. “Asymptotically Optimal Depth Fermionic Permutation on 2D Grid Quantum Architecture without Ancillas.” arXiv preprint arXiv:2605.26041 (2026).
- Kun Liu, **Shifan Xu**, Tomas Jochym-O’Connor, Zhiyang He, Shraddha Singh, and Yongshan Ding. “In-Situ Simultaneous Magic State Injection on Arbitrary CSS qLDPC Codes.” arXiv preprint arXiv:2604.05126 (2026).
- Adrian Harkness, Shuwen Kan, Chenxu Liu, Meng Wang, John M. Martyn, **Shifan Xu**, Diana Chamaki, Ethan Decker, Ying Mao, Luis F. Zuluaga, Tamás Terlaky, Ang Li, and Samuel Stein. “FTCircuitBench: A Benchmark Suite for Fault-Tolerant Quantum Compilation and Architecture.” arXiv preprint arXiv:2601.03185 (2026).
- D. K. Weiss, **Shifan Xu**, Shruti Puri, Yongshan Ding, and Steven M. Girvin. “Faulty towers: recovering a functioning quantum random access memory in the presence of defective routers.” arXiv preprint arXiv:2411.15612 (2024).

MANUSCRIPT IN PREPARATION

- Shifan Xu**, Connor T. Hann, Nathan Wiebe, Steven M. Girvin, and Yongshan Ding. “Error Corrected QRAM Queries with One-Fault-Tolerant CSWAP Gates.” 2026.

Research Software & Tutorials

PennyLane / Xanadu collaboration. Contributed QRAM implementations to PennyLane and coauthored the public tutorial below, comparing select-only, bucket-brigade, and hybrid QRAM constructions through executable examples and resource analysis. PennyLane 0.44 release announcement.

Shifan Xu, David Ren, Marcus Edwards, and Yongshan Ding. *Intro to quantum random access memory (QRAM)*. PennyLane Demos, August 30, 2026. Tutorial and code.

QRAM simulators. Developed simulation software for quantum random-access memory architectures.

Technical Skills

Programming languages: Python, C.

Research tools: QRAM simulators; quantum circuit simulation and resource estimation with PennyLane.

Awards & Fellowships

- 2020 **ZhengGang Fellowship**, Nanjing University
- 2019 **People Scholarship**, Nanjing University
- 2019 **Outstanding Prize in Taiwan College Physics Experiment Tournament (TCPT)**, National Tsing Hua University
- 2018 **Outstanding Prize in Top-notch Talent Plan**, Nanjing University

Teaching Experience

TEACHING ASSISTANTSHIPS

Yale University

New Haven, CT, USA

GRADUATE TEACHING ASSISTANT

Fall 2026

- CPSC 4310/5310: Computer Music: Algorithmic and Heuristic Composition
- Instructor: Scott Petersen

Yale University

New Haven, CT, USA

GRADUATE TEACHING ASSISTANT

2022 – 2023

- CPSC 4470/5470: Introduction to Quantum Computing
- Instructor: Prof. Yongshan Ding

GUEST LECTURES

Mar. 2026. *Introduction to Quantum Query Architecture*. CPSC 4490/5490: Quantum Information Systems, Yale University. Host: Prof. Yongshan Ding.

Feb. 2025. *Introduction to Quantum Query Architecture*. CSCI 4965, Rensselaer Polytechnic Institute. Host: Prof. Zhiding Liang.

Mentoring

Alvin Lu

New Haven, CT, USA

UNDERGRADUATE STUDENT, YALE UNIVERSITY

2024 – Present

- Mentored research on high-bandwidth shared quantum memory, leading to the coauthored paper *Fat-Tree QRAM* at ASPLOS 2025.

Presentations

INVITED TALKS

Apr. 2026. *Distilling Magic States in the Bicycle Architecture*. Invited talk hosted by the Xanadu team.

Spring 2024. *Architectural Improvements in Quantum Random Access Memory*. Invited talk at Prof. Liang Jiang's Group Meeting, University of Chicago.

Fall 2023. *Systems Architecture for Quantum Random Access Memory*. Invited talk at QAISG's Quantum Machine Learning Seminar, Centre for Quantum Technologies, National University of Singapore.

Professional Service

PROGRAM COMMITTEE

2024. Quantum Computing Applications and Systems Workshop, co-located with the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD). New Jersey, USA.

JOURNAL REFEREEING

Physical Review Research

Physical Review A (**PRA**)

PRX Quantum

Physical Review Letters (**PRL**)

Physical Review X (**PRX**)

IEEE Transactions on Parallel and Distributed Systems (**TPDS**)

ACM Transactions on Design Automation of Electronic Systems (**TODAES**)

ACM Transactions on Quantum Computing (**TQC**)

CONFERENCE REVIEWING

Quantum Information Processing (**QIP**)

Earlier Research Experience

Nanjing University - Physics Department

Nanjing, China

ADVISOR: DR. TIANJUE MIN

2021

- Graduation Thesis Project: "Analysis of Background Noise in $p\bar{p}$ Collision Experiments Based on the K-Nearest Neighbors (KNN) Algorithm"

UC San Diego - Department of NanoEngineering

(remote) San Diego, CA, USA

ADVISOR: DR. SHENG XU

2020

- Project: Circuit Design for Flexible Ultrasonic Array

Nanjing University - Engineering Department

Nanjing, China

ADVISOR: DR. XIN LI, DR. MINGHUI LU

2019-2021

- Project: Preparation and Design of Auxetic Metamaterials

Nanjing University - Physics Department

Nanjing, China

SUPERVISOR: PROF. SIHUI WANG

2018-2019

- Project: Metal Powder Spectra Analysis